

- 4 A particle P of mass 5 kg moves along a horizontal straight line. At time $t\text{ s}$, the velocity of P is $v\text{ m s}^{-1}$ and its displacement from a fixed point O on the line is $x\text{ m}$. The forces acting on P are a force of magnitude $\frac{500}{v}\text{ N}$ in the direction OP and a resistive force of magnitude $\frac{1}{2}v^2\text{ N}$. When $t = 0$, $x = 0$ and $v = 5$.

(a) Find an expression for v in terms of x .

[6]

[illegible]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) State the value that the speed approaches for large values of x . [1]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....